ASTANA IT UNIVERSITY

SEPTEMBER 22, 2026

W1

Weekly Record

BUSINESS ADMINISTRATION
CARREFOUR AND MAGNUM CASH&CARRY

PREPARED BY
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Business context and AI opportunities

Weekly Record 1 • Carrefour and Magnum Cash&Carry

1 Concept

Businesses combine natural resources, labour, capital, entrepreneurship and knowledge to create value. Retailers offer availability and convenience. Economic systems organise resources differently; inflation, employment and growth influence business decisions. Supply, demand and competition affect prices [1].

Concept Check, section 1.1: how do revenue, costs and profit differ? When a shopper buys bread at Magnum, the money received is revenue. The shop pays for the bread, wages and rent. Profit is what remains after all costs. Higher sales alone do not establish higher profit.

2 Application

Company A is Carrefour SA, an international retailer with franchised operations: 15,415 stores under its banners at 30 June 2026 [2]. Company B is Magnum Cash&Carry, serving households and businesses in Kazakhstan. Magnum reported over 200 stores on 1 July 2026 [3]. Carrefour provides documented practices to compare; Magnum provides the local setting. The question is whether assisted ordering or search could improve service at an acceptable cost.

Table 1 Environment and AI opportunity map

Factor	Effect on Magnum	Evidence	Possible AI use and control
Economic	Affordability, costs and employment affect demand.	BNS, 1 Sep. 2026: annual inflation 9.8%, Aug. 2026/Aug. 2025. BNS, 14 Aug. 2026: real GDP +4.1%, H1 2026/H1 2025, flash estimate. BNS, 13 Aug. 2026: unemployment 4.5%, Q2 2026. [4], [5], [6]	Forecast sales; category managers verify errors and approve orders. Retain manual ordering.
Political and legal	Personal data require privacy controls.	Kazakhstan Personal Data Law, 21 May 2013, No. 94-V, art. 5: lawfulness, confidentiality, security. Checked 11 Sep. 2026. [7]	Assist rule searches; legal staff verify obligations. Use approved or aggregate data.
Technological	Online shopping encourages better search and availability.	BNS, 25 May 2026: e-commerce was 14.3% of total retail trade in 2025, including marketplaces; not grocery-specific. [8]	Pilot catalogue search. Verify product facts; retain ordinary search and staff support.
Competitive	SMALL competes in Kazakhstan grocery retail.	SMALL, Small Tech, undated; accessed 11 Sep. 2026: retail network and technology infrastructure. AI deployment unconfirmed. [9]	Compare public offers; staff check matching products and dates.
Demographic and social	Market size does not establish preferences or trust.	BNS, 1 Sep. 2026: population 20,604,819 as of 1 Aug. 2026. [10]	Analyze aggregate demand; test language preferences. Keep human complaint handling.
Global supply network	Import prices may affect procurement, but do not prove delays.	BNS, Statistics of prices, accessed 11 Sep. 2026: import-price index 101.6%, Jun. 2026/Jun. 2025 (+1.6%). Publication date unconfirmed. [11]	Flag invoice changes; procurement verifies alerts and approves decisions.

Source: the dated publications cited in each row. Effects and proposed AI uses require local testing.

3 Verification and AI record

Source checks with Codex matched the periods and values in the map: inflation, real GDP, unemployment, e-commerce, population and import prices. The Carrefour count includes franchise markets; Magnum's count is a company statement. The e-commerce figure includes marketplaces. The import index of 101.6% means a 1.6% increase; its access date is used because the publication day is unspecified. These checks and source locations are recorded in the Evidence Folder.

The inflation evidence makes affordability relevant, but store sales and shopper feedback are needed to identify which products people will buy. GPT-6 Astra in the ChatGPT app helped with requirements and wording. Codex helped find and check sources, analyse course materials, draft revisions and format the document.

4 Transfer

Carrefour launched Hopla in France on 8 June 2023 [12]. Magnum could test whether Kazakh and Russian questions lead to available catalogue items, with staff help when search fails. Forecasting needs a separate comparison against current ordering, using errors and stockouts. Managers should retain order approval and complaints. Expansion depends on benefits covering costs, protected customer data and a usable manual process.

5 Evidence

The Evidence Folder records the company scale statements, Kazakhstan indicators and Carrefour's Hopla announcement, with dates, links and claim checks. Source numbers match the Evidence Folder.

6 Capstone link

This record develops Capstone section 1, Business context, and sets the questions for the comparison.

References

[1] L. J. Gitman et al., "Introduction to Business," 2nd ed., OpenStax, 2026.

[2] Carrefour, "Q2 2026 sales and H1 2026 results," Jul. 23, 2026.

[3] Magnum Cash&Carry, "How to get a job at Magnum," Jul. 1, 2026.

[4] Bureau of National Statistics of Kazakhstan, "Inflation in the Republic of Kazakhstan August 2026," Sep. 1, 2026.

[5] Bureau of National Statistics of Kazakhstan, "GDP for January to June 2026 flash estimate," Aug. 14, 2026.

[6] Bureau of National Statistics of Kazakhstan, "Labour market in the Republic of Kazakhstan Q2 2026," Aug. 13, 2026.

[7] Republic of Kazakhstan, "On Personal Data and their Protection," May 21, 2013, Law No. 94-V, art. 5.

[8] Bureau of National Statistics of Kazakhstan, "E-commerce in the Republic of Kazakhstan 2025," May 25, 2026.

[9] SMALL, "Small Tech," n.d.

[10] Bureau of National Statistics of Kazakhstan, "Demographic statistics," Sep. 1, 2026 update.

[11] Bureau of National Statistics of Kazakhstan, "Statistics of prices," n.d.

[12] Carrefour, "Carrefour integrates OpenAI technologies and launches a generative AI powered shopping experience," Jun. 8, 2023.

ASTANA IT UNIVERSITY

SEPTEMBER 22, 2026

W2

Weekly Record

BUSINESS ADMINISTRATION
CARREFOUR AND MAGNUM CASH&CARRY

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Ethics and stakeholders

Weekly Record 2 • Carrefour and Magnum Cash&Carry

1 Concept

Ethics concerns right and wrong. Utilitarianism weighs overall consequences, while duties and rights protect people whose interests might otherwise be overlooked. Managers influence behaviour through their example, training and reporting arrangements [1].

CSR includes economic, legal, ethical and philanthropic responsibilities. These operate together [13]. Concept Check, section 2.3: give an example of legal but irresponsible behaviour [1]. A lawful promotion could encourage customers to buy food they cannot use. Donations would not remove that problem.

Comparative advantage means specialising where opportunity cost is lower. Imports come from abroad; exports go abroad. Trade balance is exports minus imports. Currency depreciation, transport costs and trade barriers can make foreign purchases more expensive [1].

2 Application

Company B is Magnum Cash&Carry; Company A is Carrefour. Magnum's history records entry into Uzbekistan in 2021 [14]. This establishes regional experience, not export volumes.

The proposed use is catalogue search. Magnum Club's policy lists purchase histories and location data [15], creating a tension between useful recommendations and privacy. My priorities are customers, who bear direct harm; regulators, who influence permitted processing; and employees, who correct mistakes. Suppliers need fair visibility. Owners need a viable trial, while communities bear waste consequences.

Table 2 Stakeholder and AI ethics map

Stakeholder	Wants and risks	AI issue	Magnum obligation	Carrefour evidence	Transfer verdict
Customers	Affordable, accurate choices; unsuitable suggestions.	Wrong products or language bias.	Check stock and labels; test both languages; retain staff help.	Hopla links budget questions to product search [12].	PILOT: compare successful searches.
Employees	Manageable work; extra corrections.	Fluent answers can hide errors.	Train staff, log corrections and assign a service manager.	AI supports product descriptions; staff workload is unreported [12].	ADAPT: keep staff review.
Suppliers	Fair visibility; inaccurate product claims.	Generated descriptions may misrepresent goods.	Check descriptions with suppliers and allow corrections.	Carrefour generated descriptions for its own brands [12].	PILOT: verify supplier product facts.
Regulators	Accountability; privacy violations.	Customer data may reach external providers.	Review data flows, restrict access and retain records.	Carrefour describes Azure privacy features, not an independent audit [12].	ADAPT: assess Kazakhstan requirements.

Source: Carrefour's launch announcement [12]. The Magnum controls and transfer decisions are proposals.

3 Verification and AI record

Checks with Codex on 20 September 2026 matched the Uzbekistan entry year to Magnum's history [14], data categories to policy section 4.3 [15], and Hopla's launch to Carrefour's release of 8 June 2023 [12]. The release gives no measured error rate or staff savings. I therefore removed implied proof of effective oversight. NIST supports named responsibilities and training [16]; it does not certify Magnum.

The evidence refines my initial view: faster service matters only if shoppers can rely on it. Codex helped analyse the textbook and slides, check sources, draft the stakeholder analysis and edit this record. The checks distinguish company announcements from measured performance.

4 Transfer

PILOT catalogue search after privacy review. The digital team should test Kazakh and Russian questions against available products, comparing errors and staff correction time with ordinary search. A service manager should resolve complaints and pause the trial after harmful recommendations. DELAY purchase-history personalisation until its purpose and benefit are demonstrated.

Carrefour also proposes recipes using leftover ingredients [12]. Magnum could adapt this idea to products approaching expiry, with stock and safety checks. Store managers should compare discarded food per unit sold before and during the trial, linking it to SDG 12.3 [17]. This is a proposed measure, not a reported result.

5 Evidence

Carroll [13], NIST [16] and UN Goal 12 [17] extend the Evidence Folder. Company sources are reused. Course material: Chapter 2 and 3 slides and BA Week 2 Practice 2, slides 7–16.

6 Capstone link

This record develops Capstone section 2, Environmental and stakeholder analysis, and supports Business Report section 2.

References

[1] L. J. Gitman et al., “Introduction to Business,” 2nd ed., OpenStax, 2026.

[12] Carrefour, “Carrefour integrates OpenAI technologies and launches a generative AI powered shopping experience,” Jun. 8, 2023.

[13] A. B. Carroll, “Carroll’s pyramid of CSR: taking another look,” Int. J. Corporate Social Responsibility, vol. 1, art. 3, 2016.

[14] Magnum Cash&Carry, “About us,” n.d.

[15] Magnum Cash&Carry, “Magnum Club personal data privacy policy,” n.d.

[16] NIST, “Artificial Intelligence Risk Management Framework AI RMF 1.0,” NIST AI 100-1, Jan. 2023, GOVERN 2.1–2.3.

[17] United Nations, “Goal 12,” n.d., target 12.3.

ASTANA IT UNIVERSITY
SEPTEMBER 22, 2026

W3

Weekly
Record

BUSINESS ADMINISTRATION
CARREFOUR AND MAGNUM CASH&CARRY

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Ownership and business model

Weekly Record 3 • Carrefour and Magnum Cash&Carry

1 Concept

Ownership affects control, finance and risk. A sole proprietor has direct control but unlimited personal liability. Partners share resources and decisions; liability depends on the form. A corporation is a separate legal entity, with ownership held through shares. Entrepreneurs seek opportunities under uncertainty, while intrapreneurs develop ideas within an existing organisation [1].

Concept Check, section 5.1: what differentiates an entrepreneur from a small-business owner? Entrepreneurship emphasises innovation and growth; an owner may instead prioritise a stable operation. At Magnum, an employee proposing a forecasting trial would be acting as an intrapreneur.

2 Application

Carrefour SA operates an international retail model that includes franchises [2]. Magnum identifies itself as an LLP [15]. Kazakhstan’s partnership law generally limits participants’ risk to their contributions, subject to exceptions, and makes their general meeting the highest governing body [18]. This legal form does not reveal Magnum’s beneficial owners or its internal approval limits.

Magnum creates value by making goods available to households and business customers [14]. Sales generate revenue; purchases, wages, premises and logistics consume it. Forecasting could reduce tied-up stock and waste, while catalogue search could make shopping easier. Both also add software and staff costs. Magnum Club already provides digital loyalty interaction [15], although the policy does not establish AI deployment.

Table 3 AI Business Model Stress Test

Model element	Carrefour evidence	Stress on Magnum	Verdict	Local condition and test
Customer search	Hopla and later shopping services link questions to products. [12], [19]	Recommendations may point to unavailable goods.	ADAPT	Test Kazakh and Russian questions against the local catalogue. Compare successful searches; keep staff help.
Stock planning	SAS describes AI for demand forecasts and ordering. [20]	Stock errors or promotions make forecasts unreliable.	PILOT	Compare stockouts, waste and correction time with current ordering. Category managers approve orders.

Model element	Carrefour evidence	Stress on Magnum	Verdict	Local condition and test
Supplier relationships	Ethics charter covers conduct and reporting. [21]	Written commitments may not change purchasing.	ADOPT	Apply accountability through existing Magnum rules. [22] Track resolution and repeated complaints.
Loyalty offers	Later services use purchase histories. [19]	Extra data may add intrusion and cost without enough benefit.	DELAY	Review purpose, access and customer choices. Demonstrate value beyond existing Magnum Club offers. [7], [15]
Order approval	A proposed extension beyond the documented comparison.	Automatic commitments could magnify mistakes.	REJECT	Keep a named manager and manual ordering for this trial. This is the proposed Magnum boundary.

Source: company and vendor publications cited in the table. Verdicts and test conditions are proposed for Magnum.

Changed assumption: the trial finds that recorded stock differs from shelf stock during promotions. The forecasting verdict changes from PILOT to DELAY: category managers first reconcile stock and promotion records, then retest against current ordering. Search remains limited to items with checked availability. Expansion requires savings from fewer lost sales and less waste to exceed software, integration, training and correction costs. These are decision criteria, not measured Magnum results.

This test supports Capstone section 3 and Business Report section 3. It narrows the investment case: improve availability with accountable staff before using more customer data. Preparation and source checks were supported by Codex.

References

[1] L. J. Gitman et al., “Introduction to Business,” 2nd ed., OpenStax, 2026.

[2] Carrefour, “Q2 2026 sales and H1 2026 results,” Jul. 23, 2026.

[7] Republic of Kazakhstan, “On Personal Data and their Protection,” May 21, 2013, Law No. 94-V, art. 5.

[12] Carrefour, “Carrefour integrates OpenAI technologies and launches a generative AI powered shopping experience,” Jun. 8, 2023.

[14] Magnum Cash&Carry, “About us,” n.d.

[15] Magnum Cash&Carry, “Magnum Club personal data privacy policy,” n.d.

[18] Republic of Kazakhstan, “About limited and additional liability partnerships,” Apr. 22, 1998, Law No. 220-I, arts. 2 and 41.

[19] Carrefour, “Carrefour devient le premier retailer européen à proposer de faire ses courses sur ChatGPT,” Mar. 26, 2026.

[20] SAS, “Carrefour s’appuie sur l’Intelligence Artificielle de l’éditeur SAS pour optimiser la gestion de sa supply chain et réduire le gaspillage des produits alimentaires,” Jan. 14, 2019.

[21] Carrefour, “Carrefour Group Suppliers Ethics Charter,” 2024 edition.

[22] Magnum Cash&Carry, “Corporate rules,” n.d.

Activity 3 AI Business Model Stress Test

Kniazev Daniil | AIB-2602 | 30 September 2026

Company B: Magnum Cash&Carry Company A: Carrefour

Magnum should test forecasting and adapt catalogue search, retaining staff approval. The table separates existing activities from proposed AI uses and expected benefits.

Element	**Company B now**	**AI opportunity / change**	**Risk / VUCA pressure**	**Company A evidence**	**Verdict + why**
Customer / need	Magnum serves household shoppers and business buyers through retail and wholesale activities.	Help shoppers find available products through questions in Kazakh or Russian, reducing search effort.	Uncertainty: shoppers may prefer ordinary search. Language errors could suggest unsuitable products.	Carrefour, 8 Jun. 2023: Hopla connects questions about budgets, dietary needs and meals to product search.	**ADAPT**: use Magnum's catalogue and both languages. Test successful searches and retain staff help.
Value proposition	Stores provide access to goods. Magnum Club adds bonuses, discounts and offers based on customer interests.	Offer relevant basket suggestions; using purchase histories could extend personalisation beyond the shopper's current request.	Ambiguity: additional data may not improve relevance enough to justify privacy concerns.	Hopla responds to preferences supplied by shoppers. This supports useful suggestions without establishing a need for Magnum's purchase histories.	**DELAY** additional personalisation: establish purpose, privacy controls and benefits beyond existing offers. Start with the shopper's stated needs.
Delivery / process	Retail and wholesale operations require replenishment. The sources do not establish Magnum's forecasting method or stock-data quality.	Forecast demand by product and store; suggest quantities that category managers review before ordering.	Volatility: promotions or delivery changes can quickly alter demand and stock needs.	SAS, 14 Jan. 2019: Carrefour selected SAS Viya to improve forecasts and supplier ordering using store, warehouse and online data.	**PILOT** one category against current ordering. Compare stockouts and waste; retain manual approval because local accuracy is unproven.
Resources / activities	Magnum has store and commercial teams and a loyalty application. These sources do not establish AI capabilities.	Draft catalogue descriptions and support purchasing documents, while staff verify product facts and decisions.	Complexity: product data, suppliers, staff skills and software must work together. Errors add correction work.	Carrefour, 8 Jun. 2023: AI enriches product descriptions and supports tender drafting and quote analysis for non-retail purchasing.	**ADAPT** assistance to local products and workflows. **REJECT** autonomous approval for this trial; managers remain accountable.

Element	Company B now	AI opportunity / change	Risk / VUCA pressure	Company A evidence	Verdict + why
Revenue / value capture	Retail and wholesale sales generate revenue. Profit depends on covering goods and operating costs; loyalty discounts affect margins.	Reduce lost sales and waste through better availability. Deduct software, integration, training and correction costs from expected benefits.	Ambiguity: higher sales may reflect promotions rather than AI, and may not mean higher profit.	SAS links Carrefour's forecasting project to availability and reduced overstock. The announcement gives no measured return applicable to Magnum.	**PILOT** the business case. Finance and category managers should compare net benefits with current practice before expansion.

Sources: Carrefour, Magnum Cash&Carry and SAS. Proposed uses and verdicts are the analysis for Magnum.

Changed assumption

Suppose a promotion reveals that recorded stock differs from shelf stock. Forecasting changes from PILOT to DELAY until category managers reconcile the records. Search must then show only checked availability, staff need time to correct data, and the financial test must include that work. Retest against current ordering before expansion. This is a stress scenario, not a reported Magnum incident.

Sources

Sources checked 30 September 2026. Company and vendor announcements establish the examples; Magnum's results require local testing.

Carrefour. Carrefour integrates OpenAI technologies and launches a generative AI-powered shopping experience. 8 June 2023.

Magnum Cash&Carry. About us. undated.

Magnum Cash&Carry. Magnum Club personal data privacy policy. undated. Sections 2.4 and 4.3.

SAS. Carrefour uses AI to improve its supply chain and reduce food waste. 14 January 2019. Translated title; official SAS copy of the same release.

Weekly Record 4

Kniazev Daniil | AIB-2602 | 30 September 2026

Management, responsibility and human–AI task allocation
Carrefour and Magnum Cash&Carry

1 Concept

Management connects planning, organising, leading and controlling. Setting a target is only the first step: managers must provide resources, explain responsibilities and respond when results fall short. Chapter 7 distinguishes authority from responsibility and explains how reporting relationships support coordination [1].

Concept Check, section 7.4: how does the chain of command clarify reporting relationships? For the proposed Magnum trial, staff would report stock discrepancies to the category manager. The manager could obtain IT support without passing the purchasing decision to the software. Participative leadership would let store employees challenge forecasts using what they see on the shelves [1].

2 Application

Magnum’s retail and wholesale activities make replenishment a relevant process [14]. I propose a six-month forecasting trial in one store and product category, with the category manager accountable for ordering. Week 3 identified inaccurate promotion and stock records as a reason to delay the trial. Those records must therefore be checked first.

The matrix separates producing a recommendation from committing money. It describes proposed roles; public company material does not establish Magnum’s current allocation of these tasks.

Category	**Task**	**Reason and control**
Human Only	Approve the final order and handle exceptional commitments.	The category manager checks stock, shelf life and budget, records the decision and remains accountable [1], [16], [22].
Human + AI	Interpret promotion effects and unusual demand.	AI highlights patterns; staff add local knowledge and challenge misleading suggestions. Store feedback informs the manager’s decision [1].
AI Executes with Human Review	Produce demand forecasts and draft order quantities.	Carrefour’s forecasting example supports testing this use [20]. The manager checks inputs, unusual quantities and supplier limits before accepting or changing each order.
Automated	Check missing fields and timestamps; record versions and approvals.	These are fixed rules, not AI purchasing decisions. Failed checks block the draft and notify staff. Logs make errors traceable [16].

If records disagree or a forecast fails, staff stop the affected recommendation and alert the category manager. Unresolved supply problems go to the supply-chain lead; technical faults go to IT. Staff use a physical stock count and verified sales to prepare a manual order for the same manager’s approval. Training should include rejecting a poor forecast and practising this fallback.

3 Verification and AI record

The course text was checked for management functions, participative leadership and reporting relationships [1]. NIST supports explicit responsibilities and staff training [16]. The SAS announcement describes Carrefour’s forecasting work, but it does not establish Magnum’s performance or guarantee savings [20]. GPT-6.1 Sol helped find these sources, and draft the matrix.

The important distinction is authority: automating calculations does not mean delegating the right to place an order.

4 Transfer

PILOT Carrefour's forecasting approach after reconciling local data [20]. Adapt it to Magnum's category, promotions and supplier lead times. Keep final approval with a named manager. Do not expand if staff cannot explain the forecasts, correct errors or operate the manual process.

5 Evidence

The Evidence Folder links the textbook [1], Magnum's company description [14], NIST [16] and the SAS announcement [20] to their specific claims. The proposed management roles are an application of these sources, not a published Magnum organisation chart.

6 Capstone link

This record supplies the human–AI operating model for Assignment 2 and the capstone.

References

Source numbers and full links are in the cumulative Evidence Folder.

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. OpenStax, 2026, secs. 6.1, 6.4–6.5 and 7.4–7.5.

[14] Magnum Cash&Carry, "About us," n.d.

[16] NIST, "Artificial Intelligence Risk Management Framework AI RMF 1.0," NIST AI 100-1, Jan. 2023, GOVERN 2.1–2.3.

[20] SAS, "Multinational Carrefour relies on SAS AI to optimize supply chain management and reduce food waste," Jan. 14, 2019.

[22] Magnum Cash&Carry, "Corporate rules," n.d.

Weekly Record 5

Kniazev Daniil | AIB-2602 | 30 September 2026

Replenishment, controls and performance
Carrefour and Magnum Cash&Carry

1 Concept

Operations management coordinates people, materials and information to deliver a useful service at an acceptable cost. Resource planning balances availability against the cost of holding stock. A process map shows where work passes between people and where a delay or error can affect the whole service [1].

Concept Check, section 10.4: how is technology used in resource planning? It can combine sales and stock data to support purchasing. The benefit depends on accurate inputs and workable decisions. Faster calculations alone will not fill an empty shelf, and a forecast should not become a purchase without review.

2 Application

The process is replenishment of one product category in one Magnum store. Its retail activities and product acceptance rules establish the context [14], [22]. The current-state sequence below is an evidence-informed baseline to confirm with staff; it is not an observed account of Magnum's internal procedures. The proposed bottleneck is reconciling stock and promotion information before deciding quantities.

Step	Current state to confirm	Proposed future state
1	Check sales and stock.	Automated: validate sales and stock inputs; block failed checks.
2	Estimate demand and order quantity using the existing method.	AI Executes with Human Review: forecast demand and draft quantities.
3	An authorised manager approves the order.	Human + AI: staff interpret promotions and supplier constraints. Human Only: the category manager checks exceptions and approves or changes the order.
4	Send the order and receive goods.	Send only the approved order and receive goods.
5	Check shelves and record results.	Check shelf availability and costs. Automated logs preserve inputs, revisions and approvals for review.

If records disagree or a forecast fails, staff stop the affected recommendation and alert the category manager. Unresolved supply problems go to the supply-chain lead; technical faults go to IT. Staff prepare a manual order from verified sales and a physical stock count. The category manager approves it and authorises any restart after correction. No unapproved order is sent [16].

The following six-month pilot targets are proposals, not Magnum results. Establish a baseline before starting, use the same products and checking schedule, and compare results with the existing ordering method.

Measure	Definition and proposed target
Shelf availability	Available product checks ÷ all scheduled checks × 100. Target: at least 98%.
Forecast error (WAPE)	Sum of absolute forecast errors ÷ sum of actual units sold × 100. Target: a 10% relative reduction from baseline.
Cost and waste checks	Proposed reductions of 10% each in expired/damaged stock value and staff cost per order. Include software, training and correction costs when assessing net benefit.

Lost sales during stockouts can conceal demand, so WAPE must be interpreted alongside shelf availability. The category manager should review exceptions daily and trends weekly, using store feedback to explain unusual results [1].

Carrefour's documented forecasting initiative makes PILOT a defensible transfer decision [20]. Week 3's stock-data problem remains a stop condition. Expand only if availability reaches the target, forecast error improves, net benefit remains positive and staff can use the fallback. This map provides the operations argument and on-screen evidence for Assignment 2.

References

Source numbers and full links are in the cumulative Evidence Folder.

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. OpenStax, 2026, secs. 6.5 and 10.1, 10.4–10.5.

[14] Magnum Cash&Carry, "About us," n.d.

[16] NIST, "Artificial Intelligence Risk Management Framework AI RMF 1.0," NIST AI 100-1, Jan. 2023.

[20] SAS, "Multinational Carrefour relies on SAS AI to optimize supply chain management and reduce food waste," Jan. 14, 2019.

[22] Magnum Cash&Carry, "Corporate rules," n.d.

Weekly Record 6

Kniazev Daniil | AIB-2602 | 17 October 2026

Customer value, pricing and promotion
Carrefour and Magnum Cash&Carry

1 Concept

Marketing starts from customer value: the benefits a customer receives compared with the price and effort of buying. A firm combines product, price, place and promotion to deliver that value to a chosen target market [1]. Pricing decides how much of that value the firm captures, and promotion informs, persuades and reminds; sales promotion adds a short-term reason to buy now [1].

Concept Check, section 11.9: what is the difference between penetration pricing and price skimming? Penetration sets a low price to win market share quickly; skimming starts high and lowers the price later. A mature grocery category leaves little room to skim, so value depends on clear, credible offers.

Concept Check, section 12.8: how does sales promotion differ from advertising? Advertising builds awareness over time through paid media; sales promotion offers a temporary incentive, such as a discount, coupon or bonus, to trigger an immediate purchase. Magnum's weekly Пайда programme is a sales promotion.

2 Application

Magnum's target segment is budget-conscious households doing a weekly family shop. Annual inflation was 9.8% in August 2026 [4], and Magnum Club reports more than 2 million members [24]. Пайда listed 51 products for 15–21 October, discounted by 18–53% [23]. Of the six products reduced by 50% or more, four were household or personal-care items; staples in the list (eggs, bananas, chicken fillet) were reduced by only 18–22% [23].

The friction point lies between consideration and purchase. The Пайда page attracts attention, but its "up to 60%" promise was not met that week, and it offers no store link or Magnum Go button [23], [25]. The Week 6 artifact therefore compared two ways of writing the same weekly message from the same 51-product list.

Criterion	**Human-only baseline (Magnum's published text)**	**AI-supported draft (Claude, 17 Oct)**
Prices and dates match the catalogue	Met: dates correct	Met: checked against the saved list
One concrete offer	Not met	Met: Persil gel, 5,499 ₸ instead of 11,599 ₸
A clear next step	Not met: "just keep following Пайда"	Met: nearest store or Magnum Go
Every claim true that week	Not met: 60% claimed, 53% maximum	Met: facts from the list only
Result	1 of 4	4 of 4

Limitations: the test covered one week and one message, the baseline's authorship was assumed, and no customer response was measured. Two weaknesses in the AI draft were caught only by checking: it offered Magnum Go although Пайда price validity there was unconfirmed [25], and it picked a laundry gel rather than a food staple.

3 Verification and AI record

Prices and discounts were checked against the Пайда list saved on 17 October [23]; the maximum discount and the food and non-food split were recalculated from that file. Carrefour's figures were checked in its press releases [12], [19], and carrefour.fr was opened on 17 October to confirm that Hopla is still live [29]. Claude (Anthropic) generated the AI-supported draft that was tested and helped draft this record and its figures.

4 Transfer

ADAPT. Carrefour enriches more than 2,000 own-brand product sheets with generative AI [12], and its product pages carry this kind of marketing copy [30]. Magnum can adapt the practice for its weekly Пайда messages, but only with a fact-check gate, Kazakh-language review and no personal data in the tool. Shopping inside ChatGPT [19] is DELAY: it needs catalogue and checkout integration that this record has not assessed.

5 Evidence

The Evidence Folder links sources [4], [12], [19], [23]–[25], [29] and [30] to their claims and holds the saved 51-product list, the screenshots and the AI prompt and output.

6 Capstone link

This record supplies the customer and marketing evidence for Capstone section 5b and for Visuals 2–6 of Assignment 3.

References

Source numbers and full links are in the cumulative Evidence Folder.

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. OpenStax, 2026, chs. 11–12, secs. 11.9 and 12.8.

[4] Bureau of National Statistics of Kazakhstan, "Inflation in the Republic of Kazakhstan August 2026," Sep. 1, 2026.

[12] Carrefour, "Carrefour integrates OpenAI technologies and launches a generative AI powered shopping experience," Jun. 8, 2023.

[19] Carrefour, "Carrefour devient le premier retailer européen à proposer de faire ses courses sur ChatGPT," Mar. 26, 2026.

[23] Magnum Cash&Carry, "Magnum Пайда: Новая неделя – новые скидки," promotion page, Oct. 15–21, 2026.

[24] Magnum Cash&Carry, "Magnum Club," n.d.

[25] Magnum Cash&Carry, "Magnum Go," n.d.

[29] Carrefour, "Hopla" shopping assistant on Carrefour.fr, n.d.

[30] Carrefour, "Lait Demi-Ecrémé Grand Ouest CARREFOUR CLASSIC'," product page, n.d.

Weekly Record 7

Kniazev Daniil | AIB-2602 | 17 October 2026

Information, AI workflow and governance
Carrefour and Magnum Cash&Carry

1 Concept

Data are raw facts; information is data organised so that someone can act on it. A management information system turns transaction data into reports and decision support; its value depends on accurate inputs, clear responsibilities and protection against unauthorised access, malware and human error [1].

Concept Check, section 13.1: how do data and information differ? A list of 51 products with prices is data; a verified message telling a family which deal saves the most, until when and where to buy is information. Data become valuable when they support a better decision.

Concept Check, section 13.5: why are privacy advocates alarmed by data warehouses and data mining? Combining records from many sources can reveal detailed profiles that customers never agreed to share, and a larger store of data makes a breach more harmful.

2 Application

The decision continues Week 6: each week, which Пайда deal to feature and which message to send to which Magnum Club segment, using the pricing system, the Magnum Club CRM and app, Magnum Go, an approved AI tool and a dashboard. Internal system names are not public, so roles are assumed [23]–[25].

In the proposed workflow the AI tool receives only the public promo list and drafts three messages; segments are built inside the CRM using consent flags. A copywriter checks every price, date and claim, a native speaker checks the Kazakh text, and the brand manager approves. Personal data stay in Kazakhstan, where Law 94-V requires them to be stored [7].

Risk (likelihood / impact)	**Control and owner**	**Monitoring**
Privacy (medium / high): member data in prompts or stored abroad [7], [24]	Only the public promo list goes to AI; segments built inside the CRM. Owner: person responsible for personal data	Monthly prompt-log audit: no personal data
Reliability (medium / medium): wrong price, date or claim	Automatic price check plus copywriter sign-off; no send on mismatch. Owner: brand manager	All messages pass the check before sending
Bias (medium / medium): deepest discounts favoured over staples; weaker Kazakh copy	At least one food staple each week; native review in both languages. Owner: CRM manager	Food and non-food mix; results by language
Accountability (low / high): no record of who approved an error; tool outage	Approval log names the approver; manual template fallback. Owner: head of marketing with IT security [16]	Complete approval log; quarterly fallback drill

Uncertain cases go to the brand manager, personal-data issues to the person responsible for personal data, and AI-labelling questions to legal. If the AI tool or the check fails, the copywriter uses an approved template with the same checks; if the CRM is down, no push is sent. This fallback is not hypothetical: on 17 October Carrefour’s Hopla assistant showed a loading error twice [29].

3 Verification and AI record

The legal points were read on adilet.zan.kz: Law 94-V, articles 7, 12 and 16 [7], and AI Law No. 230-VIII, articles 17, 18 and 21 [27]. The CNIL decision was read in EDPO's unofficial English translation, which links to the CNIL original [26]. Claude (Anthropic) helped draft this record and the workflow diagram; the risk ratings are qualitative judgements. Open question: whether AI-drafted copy edited by people needs an AI label under article 21; legal should decide before a pilot.

4 Transfer

ADOPT the controls, DELAY the personalisation. Carrefour's Hopla uses purchase history but warns users not to enter personal data [29], and the CNIL fined Carrefour France €2.25 million partly for keeping the loyalty data of 28 million inactive customers for 5–10 years [26]. Magnum should first set retention limits, an AI-use notice and fast opt-out, adjusted to Kazakhstan's storage rule and AI Law [7], [27], before any personalisation with purchase history.

5 Evidence

The Evidence Folder links sources [7], [16], [23]–[27] and [29] to their claims and holds the Week 7 workflow diagram and the Hopla screenshot.

6 Capstone link

This record supplies the governance and assurance evidence for Capstone section 6 and for Visuals 7–10 of Assignment 3.

References

Source numbers and full links are in the cumulative Evidence Folder.

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. OpenStax, 2026, ch. 13, secs. 13.1 and 13.5.

[7] Republic of Kazakhstan, "On Personal Data and their Protection," May 21, 2013, Law No. 94-V.

[16] NIST, "Artificial Intelligence Risk Management Framework AI RMF 1.0," NIST AI 100-1, Jan. 2023.

[23] Magnum Cash&Carry, "Magnum Пайда: Новая неделя – новые скидки," promotion page, Oct. 15–21, 2026.

[24] Magnum Cash&Carry, "Magnum Club," n.d.

[25] Magnum Cash&Carry, "Magnum Go," n.d.

[26] CNIL, "Sanctions de 2 250 000 euros et de 800 000 euros pour les sociétés CARREFOUR FRANCE et CARREFOUR BANQUE," Nov. 26, 2020 (unofficial English translation by EDPO).

[27] Republic of Kazakhstan, "On Artificial Intelligence," Nov. 17, 2025, Law No. 230-VIII.

[29] Carrefour, "Hopla" shopping assistant on Carrefour.fr, n.d.

Weekly Record 8

Kniazev Daniil | AIB-2602 | 25 October 2026

Financial ratios and their verification
Carrefour and Magnum Cash&Carry

1 Concept

Accounting turns transactions into three statements. The balance sheet shows what a firm owns and owes on one date; the income statement shows revenue, expenses and profit over a period; the statement of cash flows shows where cash came from and where it went [1]. Ratio analysis is the framework for reading them: it expresses figures as proportions so that firms of different sizes, or one firm in different years, can be compared. Liquidity ratios ask whether short-term bills can be paid, profitability ratios how much of each sale becomes profit, activity ratios how efficiently assets are used, and debt ratios how much of the firm is financed by borrowing [1].

Concept Check, section 14.7: how can ratio analysis be used to interpret financial statements? A ratio is meaningful only against a benchmark: the same firm in another year, a competitor or a target. It shows where to ask questions rather than giving the answer. Debt ratios worry lenders because a heavily borrowed firm has less room to repay if sales fall.

Concept Check, section 14.8: how are automation and cloud tools changing accounting? Routine posting and calculation are increasingly automated, so the accountant’s work moves towards checking the data and explaining what the numbers mean for a decision.

2 Application

Magnum Cash&Carry is a private partnership and, unlike Carrefour, publishes no balance sheet or income statement that I could find. Its only public financial data are Forbes Kazakhstan’s rankings of the largest private companies, which give revenue and headcount but list net profit as “no data” [31], [32]. I therefore calculated the ratios these data allow, compared them with an AI-supported interpretation of Forbes’s Magnum profile [33], and audited every difference.

Table 1 · Magnum ratios from verified public data

Fiscal year (ranking)	Revenue, bn ₸	Employees	Revenue per employee, m ₸	Change on previous year
2023 (2024 ranking)	480.00	10,000 (rounded)	48.00	—
2024 (2025 ranking)	453.13	9,599	47.21	Revenue −5.6% nominal, −13.1% real; headcount −4.0%

Real change deflates 2024 revenue by 8.6% annual inflation [35]. Liquidity, profitability, activity and debt ratios cannot be calculated without a balance sheet and income statement.

Table 2 · Discrepancy audit: AI-supported interpretation against my calculation

AI-supported interpretation	Independent check	Outcome
Notes that figures are labelled by ranking year, but compares “between 2023 and 2025”	The 2025 ranking uses 2024 revenue and the 2024 ranking uses 2023 revenue [31], [32]	Corrected: each year moves back by one

AI-supported interpretation	Independent check	Outcome
Revenue per employee rose to 47–48 m ₸, “a sign of higher productivity”	The largest jump (+40%) falls in 2023, whose headcount is a round 10,000, the figure Magnum’s careers site uses [36]; Forbes estimates missing data [31]	Rejected for 2023; 9,599 is the only exact recent count
Revenue “slipped only about 2%”	Verified years: −5.6% nominal and −13.1% real after 8.6% inflation [35]	Qualified: the real fall is larger
Margins of 12.27% (2016) and 1.18% (2019) suggest “thin margins”	Net profit is “no data” in both recent rankings; those margins are 7–10 years old [31]–[33]	Rejected: profitability is a data gap, not a trend
Growth and revenue-per-employee arithmetic; headcount change not attributed to technology	Recalculated independently: identical to two decimals	Agreed

The audit separates two kinds of work. The arithmetic was right every time; every problem concerned the period, rounding, inflation or missing data, which required reading the source.

3 Verification and AI record

What did I check? The fiscal year behind Magnum’s revenue of 453.13 billion tenge. Against which source? The methodology of Forbes’s 2025 ranking, which ranks companies by 2024 revenue and was prepared with Adata.kz [31]. What did I find? The company profile labels the same figure “2025” [33], so a comparison with annual inflation would have been one year out. I also checked the 10,000 headcount against Magnum’s careers site, which uses the same round number for the whole company [36], and took 2024 inflation of 8.6% from the Bureau of National Statistics as reported by BES.media [35]. Magnum’s exit from Uzbekistan in 2025 [34] and Kaspi’s 2023 purchase of Magnum’s e-commerce business [31] may change what later figures cover; without statements I cannot measure that effect.

AI line: AI use for this record is declared in my A4 AI-use declaration, with what I verified and changed.

4 Transfer

ADAPT. Carrefour deployed Gemini to 250,000 employees in 2025 and uses AI agents in central functions such as sales forecasting and contract management [38]; with Vusion it aims to automate low value-added tasks and redeploy teams to customer service [37]. The logic transfers to Magnum’s head office: AI can do the calculation and the first draft. The implementation must change. Magnum should run AI on its internal ledgers rather than public estimates, put every AI figure through the four checks in Table 2 (period, rounding, inflation, scope), and have a named finance manager sign the result [16]. The scale does not transfer: Carrefour’s integrated stores employ over 300,000 people [37], against Magnum’s 9,599 [31].

5 Evidence

Added to the Evidence Folder: Forbes Kazakhstan’s 2024 and 2025 rankings [31], [32], its Magnum profile [33] and report on the Uzbekistan exit [34], BES.media’s report of 2024 inflation [35], Magnum’s careers site [36], the Carrefour–Vusion press release [37] and diginomica’s report on Carrefour 2030 [38]. The calculation and the AI interpretation are kept in the A4 appendix.

6 Capstone link

This record is developed into Capstone section 7, Business case and KPIs.

References

Source numbers and full links are in the cumulative Evidence Folder.

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. OpenStax, 2026, secs. 14.4–14.8.

[16] NIST, “Artificial Intelligence Risk Management Framework AI RMF 1.0,” NIST AI 100-1, Jan. 2023, GOVERN 2.1–2.3.

[31] Forbes Kazakhstan, “75 крупнейших частных компаний Казахстана — 2025,” Dec. 8, 2025.

[32] Forbes Kazakhstan, “75 крупнейших частных компаний Казахстана,” Dec. 11, 2024.

[33] Forbes Kazakhstan, “Magnum Cash&Carry,” company profile, n.d. Accessed: Oct. 24, 2026.

[34] Forbes Kazakhstan, “Magnum уходит из Узбекистана,” Jun. 19, 2025.

[35] BES.media, “8,6 вместо 6–8%: инфляция в Казахстане превысила ожидания,” Jan. 5, 2025.

[36] Magnum Cash&Carry, “Magnum HR,” careers website, n.d. Accessed: Oct. 24, 2026.

[37] Vusion and Carrefour, “Carrefour and Vusion join forces to deploy the smart store at scale,” press release, Feb. 18, 2026.

[38] S. Lauchlan, “How Carrefour is following Walmart’s lead as its 2030 AI-enabled transformation kicks in,” diginomica, Feb. 23, 2026.

Weekly Record 9

Kniazev Daniil | AIB-2602 | 25 October 2026

Career capability and workforce implications
Carrefour and Magnum Cash&Carry

1 Concept

Human resource management plans the people a firm needs, then recruits, selects, trains, appraises and pays them [1]. Job analysis studies what a job involves: the job description lists its tasks, and the job specification lists the skills a person needs to do them. Training and development builds those skills on or off the job, and performance appraisal checks whether the job is being done well. A knowledge worker is valued for applying what they know rather than for physical output [1]. This week's framework is the course's four capability groups: human, business, digital and meta-capabilities.

Concept Check, section 8.4: how is technology affecting off-the-job training? Online courses and simulations let people train without leaving work and repeat an exercise until they can do it. Kazakhstan's AI-Sana programme shows the scale: about 650,000 students completed a basic course in working with neural networks in its first stage [42]. When almost every graduate has taken such a course, basic AI use stops being a distinction.

Concept Check, section 9.8: how do knowledge workers differ from traditional employees? Their value depends on judgment and learning, so firms keep them through development and responsibility rather than pay alone.

2 Application

The Career Capability Portfolio maps the four capability groups against my Week 4 operating model, in which Magnum's category manager keeps order approval and AI drafts forecasts under review. Magnum separates office and store vacancies, says it trains new staff itself and runs Magnum Start for students and graduates [36]. Exposure follows international evidence: clerical work is the most exposed to generative AI, although transformation is more likely than replacement [40], and seven in ten employers rate analytical thinking as essential [41].

Table 1 · Career Capability Portfolio against the Week 4 operating model

Week 4 task	Human	Business	Digital	Meta	Value for a graduate
Human Only: approve the final order	Accountability; supplier negotiation	Budget, margin and shelf life; reading the Week 8 ratios	Forecast-error dashboards	Deciding under uncertainty and explaining it	High: authority stays with a named person [16]
Human + AI: interpret promotions and unusual demand	Store knowledge; communication	Promotion economics (Week 6: the deepest discount is not the most useful)	Challenging model output	Noticing when the model is wrong	High, and rises as AI output grows
AI Executes with Human Review: draft forecasts and quantities	Attention to detail	Supplier lead times and minimum orders	Spreadsheet and data checks (the four Week 8 checks)	Verification habit	Drafting shrinks, reviewing grows: fewer, better-prepared juniors

Week 4 task	Human	Business	Digital	Meta	Value for a graduate
Automated: data checks and approval logs	—	Process ownership	Data-quality rules and logs	Governance (Week 7 approval log)	Low in execution, moderate in design
Store customer service (where Carrefour redeploys staff [37])	Empathy; Kazakh and Russian	Service recovery	In-store systems	Resilience	Stable: human interaction shows no substitution potential [41]

Workforce implication: the most exposed entry task is drafting numbers and reports; the capabilities that keep their value are approving, interpreting and verifying. A graduate entering this model should therefore build ratio and margin literacy, data checking, and the ability to explain a decision to a manager.

References

Source numbers and full links are in the cumulative Evidence Folder.

[1] L. J. Gitman et al., Introduction to Business, 2nd ed. OpenStax, 2026, secs. 8.1, 8.4 and 9.8.

[16] NIST, “Artificial Intelligence Risk Management Framework AI RMF 1.0,” NIST AI 100-1, Jan. 2023, GOVERN 2.1–2.3.

[36] Magnum Cash&Carry, “Magnum HR,” careers website, n.d. Accessed: Oct. 24, 2026.

[37] Vusion and Carrefour, “Carrefour and Vusion join forces to deploy the smart store at scale,” press release, Feb. 18, 2026.

[40] P. Gmyrek et al., “Generative AI and jobs: A refined global index of occupational exposure,” ILO Working Paper 140, May 2025.

[41] World Economic Forum, “The Future of Jobs Report 2025,” Jan. 2025.

[42] 24.kz, “Базовый курс по ИИ завершили около 650 тысяч казахстанских студентов,” May 16, 2025.